AII. DESIGN BASIS

APROB.	VERIF.	EDICION	FECHA	DESCRIPCION DE LA EDICION
JQS	JQS/OMP	2	Sept.08	REVISADO POR CAMBIO DE RAZÓN SOCIAL
FB	ER / EL	1	Agosto 95	EMISION

JOB Nº:

CUSTOMER:

PLANT:

LOCATION:

CAPACITY:

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1. INTRODUCTION AND GENERAL INFORMATION

This Process Design Questionnaire lists information, which Intecsa Industrial requires to proceed with project design work. Having this information available at an early date will enable work to proceed rapidly and efficiently.

1.1. Correspondence

- Customer

  Company:

  Attention:

  Address:

  Telephone:

  Fax:

- Intecsa Industrial

  Attention:

  Address:

  Telephone:

  Fax:

- Others

  Attention:

  Address:

  Telephone:

  Fax:

1.2. Remarks:

2. PLANT CAPACITY

2.1. Production capacity

The plant design is based on following production capacities:

Unit	Production Capacity (t/a)

2.2. Design on-stream time

Hours per year:

2.3. Turn down ratio

Unit	% Of design capacity

3. DESIGN SCOPE AND BATTERY LIMITS

3.1. Design scope

The plant covers the following process units:

Unit nº	Description

3.2. Process battery limit conditions

If the main flow originates outside battery limits, all conditions should be specified at battery limits, with pressures at grade.

Description	Operating temp. ºC			Operating press. (kg/cm^{2}g)			Mech. design temp. ºC	Mech. design press. kg/cm^{2}g
	Min.	Norm.	Max.	Min.	Norm.	Max.		
Feedstock 1								
Feedstock 2								
Feedstock 3								
Product 1								
Product 2								
Product 3								

3.3. Battery Limits (B.L.)

The Battery Limits (B.L.) are indicated in the following drawings:

Battery Limits	Drawing Nº

4. PRODUCTS

4.1. Specification

Description	Test Method	Product 1	Product 2	Product 3

Description	Test Method	Product 1	Product 2	Product 3
Spec. gravity				
Viscosity				
At ..... ºC				
At ..... ºC				
Sulfur, %wt				
Pour point, ºC				
UOP "K" factor				
Distillation curve, ºC				
IBP				
5 %vol				
10 %vol				
30 %vol				
50 %vol				
70 %vol				
90 %vol				
95 %vol				
FBP				
Flash point, ºC				
RVP, kg/cm^2a				
Octane number				
Cetane number				

4.2. Storage

- Intermediate storage Capacity for hours/days
  Total capacity: m^3
  Nº of tanks:

- Main storage Capacity for hours/days:
  Total capacity, m^3:
  Nº of tanks:

4.3. Loading conditions

	Nº Loading Stations	Total Loading	Total Capacity	Carrying Capacity
☐ ship				
☐ ruck				
☐ railway				

Loading distance:

Loading arms to be equipped with safety devices? ☐ yes ☐ no

4.4. Remarks

5. FEEDSTOCK

5.1. Specification

Description	Test Methol	Feedstock 1	Feedstock 2	Feedstock 3

Description	Test Methol	Feedstock 1	Feedstock 2	Feedstock 3
Spec. gravity				
Viscosity				
At ..... ºC				
At ..... ºC				
Sulfur, %wt				
Pour point, ºC				
UOP "K" factor				
Distillation curve, ºC				
IBP				
5 %vol				
10 %vol				
30 %vol				
50 %vol				
70 %vol				
90 %vol				
95 %vol				
FBP				
Light Hydrocarbon :				
H_2S				
C^1				
C^2t				
C^3				
IC^4				
NC^4				
IC^5				
NC^5				

5.2. Consumption

	Feedstock 1	Feedstock 2	Feedstock 3
Normal, kg/h			
Maximum, kg/h			

5.3. Required storage capacity

Capacity, Hours/day:

Volume, m^3:

Nº of Storage tanks:

5.4. Unloading facilities

	Nº Unloading Stations	Total Unloading	Total Capacity	Carrying Capacity
☐ ship				
☐ ruck				
☐ railway				

Unloading distance:

Loading arms to be equipped with safety devices? ☐ yes ☐ no

5.5. Remarks

6. AUXILIARY MATERIALS

6.1. Specification

Description	Test Method	Auxiliary Material 1	Auxiliary Material 2	Auxiliary Material 3

6.2. Consumption

	Auxiliary Material 1	Auxiliary Material 2	Auxiliary Material 3
Normal, kg/h			
Maximum, kg/h			

6.3. Required storage capacity

Capacity, Hours/day:

Volume, m^3:

Nº of Storage tanks:

6.4. Unloading facilities

	Nº Unloading Stations	Total Unloading	Total Capacity	Carrying Capacity
☐ ship				
☐ ruck				
☐ railway				

Unloading distance:

Loading arms to be equipped with safety devices? ☐ yes ☐ no

6.5. Remarks

7. CATALYST

7.1. Specification

Description	Test Method	Catalyst 1	Catalyst 2	Catalyst 3

7.2. Consumption

	Catalyst 1	Catalyst 2	Catalyst 3
Normal, kg/h			
Maximum, kg/h			

7.3. Required storage capacity

Capacity, Hours/day:

Volume, m^3:

Nº of Storage tanks:

7.4. Unloading facilities description

_____
_____
_____
_____
____

7.5. Remarks

8. STEAM & CONDENSATE

8.1. Steam at Battery Limits

Service	Pressure, bar g				Temperature ºC				Available Flow rate t/h
	Norm.	Min.	Max.	Mecha. design	Norm.	Min.	Max.	Mechan. design	
High Pressure									
Medium Pressure									
Low Pressure									

Impurities if any:
May surplus steam at high pressure levels be consumed at low pressure levels:

☐ yes ☐ no

8.2. Steam at Boiler Plant

Service	Pressure, bar g				Temperature ºC				Available Flow rate t/h
	Norm.	Min.	Max.	Mecha. design	Norm.	Min.	Max.	Mechan. design	
High Pressure									
Medium Pressure									
Low Pressure									

Availability days/year:

8.3. Condensate at Battery Limits

Service	Pressure, bar g				Temperature ºC				Available Flow rate t/h
	Norm.	Min.	Max.	Mecha. design	Norm.	Min.	Max.	Mechan. design	
High Pressure									
Medium Pressure									
Low Pressure									

8.4. Remarks

9. WATER

Description	SERVICE						
	Curculating Cooling Water	Cooling Tower Make-up	Deminera-lized Water	Boiler Feed Water	Raw Water	Potable Water	Fire Fighting Water
Source							
Return to							
Supply pressure at B.L. bar g							
Return pressure at B.L. bar g							
Design pressure bar g							
Supply temp. max. ºC min. ºC	1)						
Max. return temperature ºC	1)						
Design temperature ºC							
Available Flow rate m^3/h							
pH							
Total hardness as $CaCO_3$ ppm							
Calcium as $CaCO_3$ ppm							
Magnesium as $CaCO_3$ ppm							
Total alkalinity as $CaCO_3$ ppm							
"P" alkalinity as $CaCO_3$ ppm							
Sulfate as SO_4 ppm							
Chloride as Cl ppm							
Silica as SiO_2 mg/l							
NH_4 mg/l							
$KMnO_4$ number							
Ferrous fe mg/l							
Dissolved Oxygen ppm							
Suspended solids mg/l							
Turbidity mg/l							
Conductivity micrS/cm							
Fouling factor	1)						
Equipment material							
Kind of water treatment							

1) for heat exchanger design

9.1. Remarks

10. COMPRESSED AIR

Description			Plant Air	Instrument Air	--------- Air
Available Flow rate,		Nm^3/h			
Pressure:	max.	bar g			
	norm.	bar g			
	min.	bar g			
	design	bar g			
Temperature	max.	bar g			
	norm.	bar g			
	min.	bar g			
	design	bar g			
Dew point		ºC			
Retention time in case of compressors trip, hours					
Storage capacity		m^3			

10.1. Remarks

11. NITROGEN

Description			Nitrogen Gas Low pressure	Nitrogen Gas High pressure	Nitrogen Liquid
Available Flow rate,		Nm^3/h			
Pressure:	max.	bar g			
	norm.	bar g			
	min.	bar g			
	design	bar g			
Temperature	max.	bar g			
	norm.	bar g			
	min.	bar g			
	design	bar g			
Purity		% vol			
Oxygen		% vol			
Storage capacity		m^3			
Service					

11.1. Remarks

12. FUEL

Description			Fuel Gas	Fuel Oil	Others
Density at ºC					
Viscosity	at ºC at ºC				
Net heating value	min. kJ/kg max. kJ/kg				
Flash point	ºC				
Molecular weight	kg/kmol				
Sulfur	% w				
Water	% w				
Vanadium	ppm w				
Ash	% w				
Sediments	% w				
H_2					
C_1					
C_2					
C_3					
C_4					
C_5					
CO					
CO_2					
O_2					
N_2					
Olefins					
Pressure:	max.	bar g			
	norm.	bar g			
	min.	bar g			
	design	bar g			
Temperature:	max.	bar g			
	norm.	bar g			
	min.	bar g			
	design	bar g			
Service					

12.1. Remarks

13. ELECTRIC POWER

13.1. Process Area

- Voltage: V ±
- Phase:
- Frenquency: Hz ±
- Receiving capacity, kVA:
- Supply system _____
_____
_____
- Number of main feeders _____
- Emergency power supply _____

max available kW, max. required kW

_____

13.2. Distribution

Voltage and phase

Service	Power Range	Voltage	Phase	Remarks
Motors	_____ kW and less			
	_____ kW to _____ kW			
	_____ kW to _____ kW			
	_____ kW and over			
Lighting				
Instruments				

13.3. Transformer type _____
_____

13.4. Special Conditions _____
_____

13.5. Fault Level _____
_____

13.6. Remarks

14. PLANT AREA

14.1. Plant Site Conditions

_____
_____
_____
_____

14.2. Site Elevation:

- Site elevation (above sea level):
- Bench mark elevation:
- Area elevation, referred to the bench mark
  - . base line level:
  - . low point existing grade:
  - . high point existing grade:
  - . high point finished grade:
  - . high point finished paving:

14.3. Soil Exploration Data:

- Soil conditions:
- Bearing value:
- Ground water level min: max:
- Ground water specification:
- Frost line depth:
- Foundation depth (below base line)
- Piling ☐ not required ☐ required
- Soil treatment ☐ not required ☐ required

14.4. Details about existing foundations and obstructions above or below base line level

_____
_____
_____
_____

14.5. Extent and Type of Paving

_____
_____
_____
_____

- Concrete Paving _____

☐ yes ☐ partly required ☐ no

- Acid proof Paving

☐ yes ☐ partly required ☐ no

14.6. Sewer Systems

Rainfall sewer	☐ yes	☐ no
Oily sewer	☐ yes	☐ no
Contaminated sewer	☐ yes	☐ no
Sanitary sewer	☐ yes	☐ no

14.7. Remarks

15. CLIMATIC DATA

15.1. Temperature and Humidity

- Mechanical design basis of equipment

  Min. Max.

  Ambient temperature, ºC:

- Minimum metal design wall temperature:

  Equipment and piping insulated, ºC:
  Equipment and piping no insulated, ºC:

- Design bases of air conditioning/ventilation:

	Summer	Winter
Outdoor temperature, ºC:		
Tolerances, ± ºC:		
Outdoor Relative humidity, %:		
Indoor temperature, ºC:		
Tolerances, ± ºC		
Indoor Relative humidity, %		

- Design basis of winterizing, ºC:

- Design basis of air cooled exchangers/evaporators:

	Air Cooler	Air Condenser	Evaporator
Ambient air temperature, ºC:			
Wet bulb temperature, ºC:			
Dry bulb temperature, ºC:			

- Design basis of electrical equipment and materials:

	Min.	Max.
Ambient temperature, ºC:		
Soil temperature, ºC:		
Indoor temperature, ºC:		
Relative humidity, %:		

- Design basis of instrument equipment and materials:

	Min.	Max.
Ambient temperature, ºC:		
Indoor temperature, ºC:		
Relative humidity, %:		

- Design basis of air blowers, air compressors, and cooling towers:

	Min.	Max.
Dry bulb temperature, ºC:		
Wet bulb temperature, ºC:		

15.2. Rainfall

Daily maximum rainfall, mm:
Design hourly rainfall, mm/hr:

15.3. Snowfall

Design snow load: kg/m^2

acc. to DIN 1055 ☐ yes ☐ no

acc. to ANSI 58.1 ☐ yes ☐ no

15.4. Wind

- Wind direction

  Prevailing direction: % of the time (or supply wind direction diagram)

- Wind load

  acc. to DIN 1055 ☐ yes ☐ no

  acc. to ANSI 58.1 ☐ yes ☐ no

  Design wind velocity, m/s:

  Design wind pressure:

Height from Ground Level, (m)	Wind Pressure (mmbar)
0	
10	
20	
30	
40	
50	
60	
70	
80	
90	
100	
100-150	
150-200	

Note: Linear interpolation for intermediate heights

15.5. Barometric Pressure

Max. barometric pressure, mbar:

Min. barometric pressure, mbar:

Design barometric pressure, mbar:

15.6. Seismic factor

acc. to DIN 1055 ☐ yes ☐ no

acc. to ANSI 58.1 ☐ yes ☐ no

15.7. Other data

(1) Thunder and lightning shall be considered? ☐ yes ☐ no

(2) Sea breeze shall be considered? ☐ yes ☐ no

(3) Sand storm shall be considered? ☐ yes ☐ no

15.8. Remarks

16. CODES AND STANDARDS

- Codes and Standards

Engineering standard	☐ II standard			
	☐ Customer's standard			
National standard	☐ DIN			
	☐			
Material specification	☐ DIN			
	☐ ASTM			
	☐			
Petroleum industry code	☐ API			
	☐			
Pressure vessels and boiler code	☐ AD			
	☐ ASME			
	☐	Heat exchanger	☐ DIN	
	☐ TEMA	Class:	☐ R	☐ C ☐ B
Pumps and compres.	☐ DIN			
	☐ API			
	☐ Manufacturer standards			
Machines	☐			
	☐			
	☐ Manufacturer standards			
Piping	☐ DIN			
	☐ ANSI			

Electrical code	☐ IEC
	☐ DE
Instrumentation	☐ IEC
	☐
Civil code	☐ DIN
	☐
Hazardous area clasif.	☐ Ex-RL
	☐ API
Tanks	☐ DIN
	☐ API
Pressure relieving devices	☐ DIN/AD
	☐ API
Fire Fighting Protection:	☐ NFPA
	☐ AFNOR
	☐ FOC
	☐
Air conditioning:	☐ ASHRAE
	☐ IT-IC
	☐

17. LAWS & REGULATIONS

The following local laws and regulations shall be observed for:

Pressure Vessels _____

Boilers _____

Buildings and Structures _____

Electrical _____

Fire fighting _____

Air conditioning _____

Water Pollution _____

Air Pollution _____

Safety _____

Sanitation _____

Aircraft Warning _____

Noise _____

Storage Tanks _____
_____
_____

18. UNITS OF MEASURE

	MKS	SI	ENGLISH
Temperature	☐ ºC	☐ ºC	☐ ºF
		☐ K	
Pressure	☐ kg/cm^2	☐ kPa	☐ psi
Mass	☐ kg	☐ kg	☐ lb
Length	☐ m	☐ m	☐ ft
	☐ mm	☐ mm	☐ in
Surface	☐ m^2	☐ m^2	☐ ft^2
Volume	☐ m^3	☐ m^3	☐ ft^3
Flowing density	☐ kg/m^3	☐ kg/m^3	☐ lb/ft^3
	☐ g/cm^3	☐ g/cm^3	
Nominal Pipe diameter	☐ mm	☐ mm	☐ in
Vessel nozzle sizes	☐ mm	☐ mm	☐ in
Flow, liquids	☐ m^3/h	☐ m^3/h	☐U.S. gal/min ☐ bpd
	☐ kg/h	☐ kg/h	☐ lb/h
Flow, gases	☐ m^3/h(1)(2)	☐ m^3/h(1)(2)	☐ Ft^3/s(3) ☐ MM ft^3/d(3)
	☐ kg/h	☐ kg/h	☐ lb/h
Heat rate	☐ MM kcal/h	☐ MW	☐ MM Btu/h
Enthalpy	☐ kcal/kg	☐ kJ/kg	☐ Btu/lb.
Power	☐ CV	☐ kW	☐ HP
Viscosity	☐ cP	☐ mPa.s	☐ cP
			☐ lbm/Ft.h
Heat transfer coefficient	☐ $kcal/hm^2$ C	☐ W/m^2 K	☐ Btu/hft^2 F
Velocity	☐ m/s	☐ m/s	☐ ft/s

Notas:

(1) Normal conditions 0 C and 101,325 KPa (1,033 kg/cm^2a)
(2) Standard conditions 15 C and 101.325 KPa (1,033 kg/cm^2a)
(3) Standard conditions 60 F and 14.696 Psia

19. NUMBERING SYSTEM

19.1. Standard Application

Equipment: ☐ II ☐ Customer

Instrumentation: ☐ II ☐ Customer

Piping: ☐ II ☐ Customer

Electrical: ☐ II ☐ Customer

19.2. Symbols and Numbering System

☐ II ☐ Customer

19.3. Drawing numbering system

Example: _____

_____

19.4. Equipment codes

ITEM	SYMBOL	ITEM	SYMBOL
Reactors, Fractionation Columns	D	Pumps, Turbines	G
Vessels, Tanks	F	Compressors, Fans, Agitators	C
Heat Exchangers, Air Coolers	E	Filters	M
Fired Heaters, Flares, Chimneys	B	Electrical Motors/Turbines	XM/XT (1)

(1) "X" indicates equipment to be driven

Example: _____

_____

_____

19.5. Piping

Example: _____

19.6. Instrumentation

Example: _____

19.7. Electrical

Example

19.8. Safety valves

Example: _____

19.9 Incoming and outgoing streams

Example: _____

19.10. Symbols and abbreviations

To be indicated and supplemented in attached PID, PFD.

20. FLUID CODES

For all process and utility lines the following fluid codes shall be used:

☐ II ☐ Customer

21. PROCESS FLOW AND PIPING AND INSTRUMENT DIAGRAMS

21.1. Process Flow Diagrams

The process flow diagrams will be prepared in accordance with Intecsa Industrial standards, with material balance information.

Depending upon the complexity of the material balance information (flow rates, composition, etc.), it may be presented on separate computer outputs rather than on the process flow diagrams.

Please state any special requirements regarding information that is to appear on the process flow diagrams.

_____
_____
_____
_____
_____

21.2. Process Piping and Instrument Diagrams (PID'S)

The process piping and instrument diagrams will be prepared in accordance with Intecsa Industrial standards. Please state any special requirements.

_____
_____
_____
_____
_____

22. BUILDINGS

22.1. Administration Building

Air conditioning: ☐ yes ☐ no

Fire Fighting Protection: ☐ yes ☐ no

22.2. Control house

Air conditioning: ☐ yes ☐ no

Fire Fighting Protection: ☐ yes ☐ no

22.3. Administration

Air conditioning: ☐ yes ☐ no

Fire Fighting Protection: ☐ yes ☐ no

22.4. Workshop

Air conditioning: ☐ yes ☐ no

Fire Fighting Protection: ☐ yes ☐ no

22.5 Laboratory

_____

_____

_____

Air conditioning: ☐ yes ☐ no

Fire Fighting Protection: ☐ yes ☐ no

22.6 Warehouse

_____

_____

_____

Air conditioning: ☐ yes ☐ no

Fire Fighting Protection: ☐ yes ☐ no

23. FIREPROFING

23.1. Skirt of Columns and Drums

23.2. Pipe Racks

23.3. Steel Structures and Steel Supports

23.4. Special Conditions

23.5. Remarks

24. PRESSURE RELIEF AND FLARE SYSTEM

24.1. Pressure Relief System

- Pressure relief valves in hydrocarbon vapor service shall discharge to:

  ☐ A closed relief header routed to a knockout drum and flare

  ☐ Atmosphere

  ☐ Standards to be considered and max. allowed output in kg/h to be stated.

  _____
  _____
  _____

- Pressure relief valves in hydrocarbon liquid service shall discharge to:

  ☐ The same header used for pressure relief valves in hydrocarbon vapor service

  ☐ A separate drum vented to the vapor header

  ☐ Other describe:

  _____
  _____
  _____

- Pressure relief valves in non-hydrocarbon service

  _____
  _____
  _____

24.2. Flare System

Type of flare:

☐ smokeless ☐ non-smokeless

24.3. Remarks

25. ANTI-POLLUTION MEASURES

25.1. Noise

_____
_____

- Plant noise level: _____
_____
_____
_____

- Community noise level: _____
_____
_____
_____

- Equipment noise level: ≤ ____ dB(A) measured at a distance of ____ m from the source

_____
_____

25.2. Water Disposal

Disposal	Allowable Contamination (mg/l)	
Oily sewers		
Chemical sewers	COD BOD	
Surface drainage		

Treated effluent water from the plant shall meet the specification as per enclosed sheet ☐

25.3. Solid Waste Disposal

Incinerable wastes: _____
_____

Non Incinerable wastes: _____
_____

25.4. Gaseous Emission

25.5. Additional Requirements

26. INSULATION

- Prefered Insulation and Covering:

	Insulation Type		Weather Covering		
Item	Block	Blanket	Mastic	Sheet Metal	Others
Columns/vessels					
Heat exchanger					
Fired heaters					
Piping					

- Insulation acc. to II Standard ☐

  Customer's Standard ☐

- Design basis of insulation

  . The equipments and piping, temperature of which is under 60 C, are not insulated. Above 60 C, personnel protection (for metallic equipments and piping) or heat conservation type insulation are used according to process requirement.

  . The equipments and pipings below 12 C are insulated where it is necessary for cold insulation or condensation prevention.

- Special Insulation (Fire, cold, condensation prevention _____
  _____

26.1. Remarks

27. PLANT LAYOUT

27.1. Basic Layout Plan

_____
_____
_____

27.2. Width of Roads

Main roads:

Secondary roads:

27.3. Spacing of Units and Equipments

_____
_____
_____

27.4. Provisions to be made for maintenance

_____
_____
_____

27.5. Equipment Transportation Limits at the Site

_____
_____
_____

27.6. North Direction

Relation between construction north and geographic north is: _____
_____

As described in drawing nº: _____
_____

27.7. Remarks

28. EQUIPMENT DESIGN

28.1. Vessels, drums, columns

- Vessels with dia. of mm will have flanged ends and handholes, all other vessels are equipped with manholes of size DN min.

- Minimum nozzle size: DN

- Sizes of drains & vents:

Vessel Volume	Vent Size	Drain Size
< 6 m^3	DN 50	DN 50
6 m^3 - 15 m^3	DN 50	DN 50
> 15 m^3	DN 50	DN 80

- Corrosion Allowances:

Carbon steel/Low alloys steel (1), mm: ☐ 3 ☐

Medium/High alloy Steel (2), mm: ☐ 1,5 ☐

Non Ferrous alloy, mm: ☐ 0,5 ☐

Alloy Linings and Cladding materials, mm: ☐ 0 ☐

(1) Carbon Steel, ½% Cr - ½% Mo, 1% Cr - ½% Mo, 1¼% Cr - ½% Cr - ½% Mo, 2¼% Cr - 1% Mo, 5% Cr - 1½% Mo.

(2) Stainless Steel, 9% Cr - 1% Mo, >9% Cr.

- Remarks

28.2. Columns internals

- Type of tray

  Preferable type of tray

  ☐ valve trays ☐ sieve trays ☐

  _____
  _____
  _____

- Tray Material _____
  _____
  _____

- Column Packings _____
  _____
  _____

- Remarks

28.3. Heat exchanger

- Air Coolers

  . Extent for the use of air cooled exchangers _____
  _____
  _____

  . Preferred tube length, mm:

  ☐ 6m ☐ 8m ☐ 10m ☐ 12m ☐_____ m

  . Overdesign:

  ☐ 5% ☐ 10% ☐15% ☐ ____ %

  . Tube thickness:

	O.D. 1"
Carbon Steel/Low Alloy Steel (9% Cr)	☐ BWG 14
	☐ BWG
Stainles Steel, Hight Alloy Steel (>9% Cr), Copper Alloys	☐ BWG 16
	☐ BWG
Others	☐ BWG

- Remarks

- Shell and Tube Units

  . Standard dimensions:

Preferred tube length	☐ 5000 mm	☐ 6000 mm
	☐ 7200 mm	☐ _____ mm
Preferred tube O.D.	☐ 20 mm	☐ 25 mm
	☐ ___ mm	

Tube thickness:

	O.D. ¾"	O.D. 1"
Carbon Steel Low Alloy Steel (9% Cr)	☐ BWG 14	☐ BWG 12
	☐ BWG	☐ BWG
Stainless Steel, High Alloy Steel (>9% Cr)	☐ BWG 16	☐ BWG 14
	☐ BWG	☐ BWG
Copper Alloys	☐ BWG 18	☐ BWG 16
	☐ BWG	☐ BWG
Other	☐ BWG	☐ BWG

  . Tube bundle: max. diameter, mm:

  max. weight, t:

- Remarks

28.4. Heaters

- Burners

  Heaters shall be provided with gas burners without provisions for future installation of oil burners ☐

  Heaters shall be provided with gas burners with provisions for future installation of oil burners ☐

  Heaters shall be provided with gas burners for onstream operation and with oil burners for start-up and stand-by purposes. ☐

- Heaters shall be provided with oil burners ☐

- Heaters shall be provided with both oil and gas burners arranged to fire either, or both fuels individually, or simultaneously at full load conditions. ☐

  _____
  _____

- A pilot burner shall be provided for each oil or combination burner, unless otherwise indicated.

  _____
  _____

- Stack height shall be ____ m. minimum above base line level.

- Remarks

28.5. Pumps and compressors

- Pumps

Pumps in process area: ☐ API 610 ☐ DIN ☐ Other

Pumps in nonprocess area: ☐ Manufacturer's standard ☐ Other

Speed: Preferable speed is _____ RPM. However _____ RPM speed can be used, unless other process requirements prevail.

- Compressors

Compressors suitable for outdoor installation ☐

Compressors to be installed inside of a compressor house ☐

- Driver

Equipment	Operating	Spare	Remarks
Pumps			
Compressors			
Blowers			
Machines			

- Spare

Equipment	Full Spare (1+1)	Hook up (2+1)	Min. Spare	No Spare
Pumps				
Compressors		----------		
Blowers		----------		
Machines				

Where process stand-by installation is not possible one spare unit shall be provided

☐

- Remarks

29. INSTRUMENTATION

29.1. Control System

Design with DCS in Control Room ☐

Conventional Design ☐

- The control panel shall be: ☐ non-graphic ☐ semi-graphic ☐ graphic
- Instruments shall be:

  Panel mounted electronic controllers ☐
  Electronic indication of temperature

  Panel mounted electronic indication and control for all loops ☐

  Pneumatic control with field mounted flow and pressure controls ☐

- The multi-point temperature indicator shall be mounted on:

  ☐ console desk ☐ panel board

- Local Control panels with:

  Electronic indication/control ☐

  Subsystems of DCS ☐

  Pneumatic indication/control ☐

  Local pneumatic controllers will be provided for subordinate control loops.

29.2. General

- Control valves in pipings up to a nominal diameter of 6" will be provided with by-passes and manual control valves if no safety function is included

  ☐ yes ☐ no

- Quick-closing valves for inflammable fluids at the battery limit

  ☐ yes ☐ no

- May trip functions act on control valves?

  ☐ yes ☐ no

- What will be the criteria for a double or multiple provision of measuring instruments/analyzers for reasons of availability?

  _____
  _____
  _____

- Instrument calibrations are to be:

Pressure	☐	bar g	☐
Temperature	☐	ºC	☐

Flow rate:

. Liquid	☐	m^3/h	☐
. Vapors	☐	Nm^3/h	☐
. Steam	☐	kg/h	☐
. Chemicals	☐	m^3/h	☐
. Water	☐	m^3/h	☐

- What will be the allowable tolerances for the measurements used for guarantee test run?.

  _____
  _____
  _____

- What additional measurements for the preparation of material balances shall be of higher accuracy?

  _____
  _____
  _____

- What calibration grade measurements will be demanded?

  _____
  _____
  _____

- To what extent shall heat exchangers be equipped with temperature points to measure their performance

  ☐ thermowells only ☐ temperature indicating point ☐

- What machines will be equipped with local control panels?

  _____
  _____
  _____

- Package units with a control system of their own which will not be integrated into the DCS:

  _____
  _____
  _____

- Machine drives that will be started from the control room:

  _____
  _____
  _____

- The guarantee values for the electric power consumption will be determined:

  ☐ for each unit ☐ for the whole plant

- In the case of a breakdown of the electric power supply system, the DCS shall remain serviceable for ..... hours.

- The following plant sections shall be equipped with Ex and TLV monitoring systems:

Ex	TLV

- Proposed suppliers:

  . Control valves

  _____
  _____
  _____

  . DCS/Panel Instruments

  _____
  _____
  _____

  . Transmitters (P-/dp-)

  _____
  _____
  _____

  . Analyzers

  _____
  _____
  _____

29.3. Instrument Tapping Sizes

Instrument type

	Instrument Connection	Piping Connection	Equipment Connection
Pressure:			
Pressure Gauge/Pressure Switch	½" male thread 1)	1" nozzle	1" nozzle
Diaphragm Gauge/Pressure Switch	2" flanged	2" nozzle	2" nozzle
Pressure Transmitter	½" (m) 1)	1" nozzle	1" nozzle
Flanged Type Press. Transm.	3" flanged	3" nozzle	3" nozzle
Remote Seal Type Press. Transm.	3" flanged	3" nozzle	3" nozzle
Temperature:			
Temperature	1" flanges 2)	1" nozzle	1" nozzle
Gauge/RTD/			
Thermocouple			
Thermowell			
Level:			
Level Gauges	1" flanged	-	1" nozzle
Magnetic Level Indicator			
Level Switch (External Float Switch,side-mounted)	2" flanged	-	1" nozzle
DP Level Transmitter	½" (F)	-	1" nozzle
Flanged Level Transmitter	3" flanged	-	3" nozzle 3)
Remote Seal Level Transmitter	2" wafer	-	3" nozzle 3)
Displacer Level Trnasmitter	2" flanged	-	2" nozzle
Purge Level Transmitter	½" (F)	-	1" nozzle
Capacitive Level Switch	1½" (M)	-	2" nozzle

1) NPT or R ½"

2) Special conditions (high pressure, high fluid velocity, highly corrosive fluid) may require special thermowells and nozzles.

3) Depending on measuring range.
(Small range: DN 4")

4) General remarks:

1" ≈ DN 25
2" ≈ DN 50
3" ≈ DN 80 etc.

Special equipment/pipe design like rubber/PTFE lining, enamel lining, plastic pipes or vessel etc. may require different nozzle design and size.